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Probabilistic Artificial Intelligence
Questions Pack

February 3, 2023

Time limit: 120 minutes

Instructions. This pack contains all questions for the final exam. It contains the questions only. Please use the accompanying answer sheet to provide your answers by blackening out the corresponding squares. As the exam will be graded by a computer, please **make sure to do blacken out the whole square and do not use ticks or crosses**. *During* the exam you can use a **pencil** to fill out the squares as well as an **eraser** to edit your answers. *After* the exam is over, we will collect the questions pack and provide you with additional 10 minutes of time to blacken out the squares on the answer sheet with a **black pen**. *Nothing* written on pages of the question pack will be collected or marked. **Only the separate answer sheet with the filled squares will be marked**.

Please make sure that your answer sheet is clean and all answers are clearly marked by filling the squares out completely. We reserve the right to classify answers as wrong without further consideration if the sheet is filled out ambiguously.

Collaboration on the exam is strictly forbidden. You are allowed a summary of *two* A4 pages and a simple, non-programmable calculator. The use of any other helping material or collaboration will lead to being excluded from the exam and subjected to disciplinary measures by the ETH Zurich disciplinary committee.

Question Types In this exam, you will encounter the following question types.

- **Multiple Choice questions with a single answer.**

  Multiple Choice questions have **exactly one** correct choice. Depending on the difficulty of the question **2, 3, or 4** points are awarded if answered correctly, and **zero points** are awarded if answered wrongly or not attempted.

- **True Or False questions.**

  Each True Or False question has a value of either **1 point** or **2 points** if answered correctly, **0 points** if answered wrongly or not attempted.

There are no negative points in this exam. Please also note that **not** all questions need to be answered correctly to achieve the best grade.

Questions in the same section might be related. In addition, some questions require information from previous parts. Hence, if you intend to skip some questions, make sure that you read all notes and details between the questions.

The notation $\mathcal{N}\left(\mu, \sigma^2\right)$ will designate a Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}\left(3, 4\right)$ designates the Gaussian distribution with mean 3 and standard deviation 2.

Good luck!

1 Probability, Bayesian linear regression

Imagine you are a gardener that wants to grow basil. Since you know that eugenol is a key volatile compound that gives basil its distinctive scent, you want to predict its production as a function of the solar radiation the plant receives. To this end, you consider the following model

$$y = wx + \varepsilon, \tag{1}$$

where $x \in \mathbb{R}$ is the solar radiation the plant is exposed to, $y \in \mathbb{R}$ is the amount of eugenol it produces, and $\varepsilon \in \mathbb{R}$ is the i.i.d. measurement noise, which has the probability density function

$$p(\varepsilon) = \begin{cases} 0.25 & \text{if } -1.5 \leq \varepsilon < -0.5, \\ 0.5 & \text{if } -0.5 \leq \varepsilon < 0.5, \\ 0.25 & \text{if } 0.5 \leq \varepsilon \leq 1.5, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

For simplicity, you consider only three possible values for w, namely $w = 0.5$, $w = 1$, and $w = 2$. Before observing any data, your expert gardening knowledge makes you believe that $p(w = 0.5) = 0.5$, $p(w = 1) = 0.25$, and $p(w = 2) = 0.25$.
Suppose you run experiments to learn more about the eugenol concentration as a function of radiation. As a result, you collect the data $x_1 = 1, y_1 = 1.2, x_2 = 2, y_2 = 1.9$ denoted by $\mathbf{X}, \boldsymbol{y}$.

Question 1 *(2 points)* What is the likelihood $p(\boldsymbol{y}|w, \mathbf{X})$ of the data set for the three values of w considered?

A $p(\boldsymbol{y}|w = 0.5, \mathbf{X}) = \frac{1}{4}, \quad p(\boldsymbol{y}|w = 1, \mathbf{X}) = \frac{1}{2}, \quad p(\boldsymbol{y}|w = 2, \mathbf{X}) = \frac{1}{4}$

B $p(\boldsymbol{y}|w = 0.5, \mathbf{X}) = \frac{1}{16}, \quad p(\boldsymbol{y}|w = 1, \mathbf{X}) = \frac{1}{4}, \quad p(\boldsymbol{y}|w = 2, \mathbf{X}) = 0$

C $p(\boldsymbol{y}|w = 0.5, \mathbf{X}) = \frac{1}{4}, \quad p(\boldsymbol{y}|w = 1, \mathbf{X}) = \frac{3}{4}, \quad p(\boldsymbol{y}|w = 2, \mathbf{X}) = 0$

D $p(\boldsymbol{y}|w = 0.5, \mathbf{X}) = \frac{1}{8}, \quad p(\boldsymbol{y}|w = 1, \mathbf{X}) = \frac{7}{16}, \quad p(\boldsymbol{y}|w = 2, \mathbf{X}) = \frac{1}{16}$

Solution:
B. Since the noise is i.i.d., the likelihood of the data set is the product of the likelihood of the single data points. For a given w, the likelihood of the data point (x, y) is given by $p(\varepsilon = y - wx)$. Therefore, the likelihoods are $p(\boldsymbol{y}|w = 0.5, \mathbf{X}) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$, $p(\boldsymbol{y}|w = 1, \mathbf{X}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, $p(\boldsymbol{y}|w = 2, \mathbf{X}) = \frac{1}{4} \cdot 0 = 0$.

Question 2 *(2 points)* How does the data change your initial belief about w? In other words, what is the distribution over w given the data set, $p(w|\mathbf{X}, \boldsymbol{y})$?

A $p(w = 0.5|\mathbf{X}, \boldsymbol{y}) = \frac{1}{3}, \quad p(w = 1|\mathbf{X}, \boldsymbol{y}) = \frac{2}{3}, \quad p(w = 2|\mathbf{X}, \boldsymbol{y}) = 0$

B $p(w = 0.5|\mathbf{X}, \boldsymbol{y}) = \frac{1}{3}, \quad p(w = 1|\mathbf{X}, \boldsymbol{y}) = \frac{1}{3}, \quad p(w = 2|\mathbf{X}, \boldsymbol{y}) = \frac{1}{3}$

C $p(w = 0.5|\mathbf{X}, \boldsymbol{y}) = \frac{1}{6}, \quad p(w = 1|\mathbf{X}, \boldsymbol{y}) = \frac{1}{3}, \quad p(w = 2|\mathbf{X}, \boldsymbol{y}) = \frac{1}{2}$

D $p(w = 0.5|\mathbf{X}, \boldsymbol{y}) = \frac{1}{3}, \quad p(w = 1|\mathbf{X}, \boldsymbol{y}) = \frac{1}{9}, \quad p(w = 2|\mathbf{X}, \boldsymbol{y}) = \frac{1}{9}$

Solution:
A. Using Bayes rule, we can derive the posterior as $p(w|\mathbf{X}, \boldsymbol{y}) = \frac{p(\boldsymbol{y}|\mathbf{X}, w)p(w)}{p(\boldsymbol{y}|\mathbf{X})}$. Since we already know the likelihood of the data $p(\boldsymbol{y}|w, \mathbf{X})$ and the prior is given, we simply need to calculate the evidence $p(\boldsymbol{y}) = \sum_{w \in \{0.5,1,2\}} p(\boldsymbol{y}|\mathbf{X}, w)p(w) = \frac{1}{16} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = \frac{3}{32}$. With this, we can compute $p(w = 0.5|\mathbf{X}, \boldsymbol{y}) = \frac{1}{32} \cdot \frac{32}{3} = \frac{1}{3}$, $p(w = 1|\mathbf{X}, \boldsymbol{y}) = \frac{1}{16} \cdot \frac{32}{3} = \frac{2}{3}$, $p(w = 2|\mathbf{X}, \boldsymbol{y}) = 0 \cdot \frac{32}{3} = 0$.

Question 3 *(4 points)* We would like to predict the probability of observing a eugenol response y^* between 2.5 and 3, when the plant is exposed to a radiation of $x^* = 3$. In other words, what is the probability $p(y^* \in [2.5, 3]|x^* = 3, \mathbf{X}, \boldsymbol{y})$?

A $p(y^* \in [2.5, 3]|x^* = 3, \mathbf{X}, \boldsymbol{y}) = \frac{1}{12}$

B $p(y^* \in [2.5, 3]|x^* = 3, \mathbf{X}, \boldsymbol{y}) = \frac{5}{24}$

C $p(y^* \in [2.5, 3]|x^* = 3, \mathbf{X}, \boldsymbol{y}) = \frac{7}{24}$

D $p(y^* \in [2.5, 3]|x^* = 3, \mathbf{X}, \boldsymbol{y}) = \frac{5}{8}$

E $p(y^* \in [2.5, 3]|x^* = 3, \mathbf{X}, \boldsymbol{y}) = \frac{7}{18}$

Solution:
B. To compute this, we want to marginalize out the weight w. In particular,

$$p(y^*|x^*, \mathbf{X}, \boldsymbol{y}) = \sum_{w \in \{0.5,1,2\}} p(y^*, w|\mathbf{X}, \boldsymbol{y}, x^*) \tag{3}$$

$$= \sum_{w \in \{0.5,1,2\}} p(y^*|x^*, w)p(w|\mathbf{X}, \boldsymbol{y}) \tag{4}$$

$$= \frac{1}{3} \cdot p(y^*|x^*, w = 0.5) + \frac{2}{3} \cdot p(y^*|x^*, w = 1). \tag{5}$$

Since y is a continuous random variable, to find out its probability to fall in the interval $[2.5, 3]$, we need to integrate its density over that interval,

$$p(y^* \in [2.5, 3]|x^* = 3, \mathbf{X}, \boldsymbol{y}) = \frac{1}{3} \cdot \int_{2.5}^{3} p(y^*|x^*, w = 0.5)dy^* + \frac{2}{3} \cdot \int_{2.5}^{3} p(y^*|x^*, w = 1)dy^* \tag{6}$$

$$= \frac{1}{3} \cdot \int_{1}^{1.5} p(\varepsilon)d\varepsilon + \frac{2}{3} \cdot \int_{-0.5}^{0} p(\varepsilon)d\varepsilon \tag{7}$$

$$= \frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{2} + \frac{2}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{3} \cdot \left(\frac{1}{8} + \frac{1}{2}\right) = \frac{5}{24}. \tag{8}$$

Question 4 *(3 points)* As you collect more data, you obtain a larger data set $\mathbf{X}', \boldsymbol{y}'$ that makes you doubt the linear relation between x and y you have considered so far. Therefore, you want to try a quadratic model, $y = w'x^2 + \varepsilon$, with ε i.i.d and following the same distribution $p(\epsilon)$ as specified above. With this assumption and the prior probabilities and likelihoods, specified in the following table

w'	0.5	1	2
$p(w')$	1/2	1/4	1/4
$p(\boldsymbol{y}'|w', \mathbf{X}')$	1/16	1/8	1/8

what is the evidence $p(\boldsymbol{y}'|\mathbf{X}')$ for this model?

A $p(\boldsymbol{y}'|\mathbf{X}') = \frac{3}{16}$

B $p(\boldsymbol{y}'|\mathbf{X}') = \frac{1}{16}$

C $p(\boldsymbol{y}'|\mathbf{X}') = \frac{1}{8}$

D $p(\boldsymbol{y}'|\mathbf{X}') = \frac{3}{32}$

Solution:
D. The model evidence is computed as

$$p(\boldsymbol{y}'|\mathbf{X}') = \sum_{w' \in \{0.5,1,2\}} p(\boldsymbol{y}, w'|\mathbf{X}) \tag{9}$$

$$= \sum_{w \in \{0.5,1,2\}} p(\boldsymbol{y}|w', \mathbf{X}) p(w') \tag{10}$$

$$= \frac{1}{2} \cdot \frac{1}{16} + \frac{1}{4} \cdot \frac{1}{8} + \frac{1}{4} \cdot \frac{1}{8} = \frac{3}{32} \tag{11}$$

1.1 Short Questions

Question 5 *(1 point)* In Bayesian linear regression, the posterior distribution over function values at test points is always Gaussian regardless of the prior distribution over the weights and the noise model.

A True B False

Solution:
False. There are many prior-likelihood combinations that yield a non-Gaussian posterior predictive distribution.

Question 6 *(1 point)* Bayesian linear regression can be applied to non-linear transformations of the input variable x.

A True B False

Solution:
True. BLR can be applied to any model of the form $y = w^\top \phi(x) + \varepsilon$.

Question 7 *(1 point)* In Bayesian linear regression, we use a single deterministic weight vector w when making a prediction y^* for a test point x^*.

A True B False

Solution:
False. We consider all the possible values of w when we marginalize it to compute the predictive distribution.

Question 8 *(1 point)* Consider the model $y = wx + \varepsilon$, where ε is i.i.d. Gaussian observation noise with mean $\mu > 0$ and standard deviation $\sigma = 1$. Any data point (x, y) with $y < 0$ has conditional likelihood equal to zero for all $w > 0$, i.e. $p(y \mid x, w) = 0$ for all $w > 0$.

A True B False

Solution:
False. The support of Gaussian random variables is the full real line. Therefore, every point has likelihood greater than 0 under this model.

2 Gaussian Processes

Suppose we are given a dataset $\mathcal{D} = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), ..., (\boldsymbol{x}_n, y_n)\}$. We assume that every label y_i is generated according to

$$y_i = f(\boldsymbol{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{12}$$

where $\boldsymbol{x}_i \in \mathbb{R}^d, y \in \mathbb{R}$ and f is an unknown function, modeled using a Gaussian process $f \sim \mathrm{GP}(0, k)$. Here k refers to the kernel or covariance function. This notation will be used across the following section.

2.1 Choice of Kernels

For the following questions, let $n = 10$, and $d = 1$. Given the dataset $\mathcal{D}$, we compute the posterior mean and 95% confidence region for four choices of k, with results displayed in Figure 1. We consider the following kernels:

- RBF Kernel: $k(x, x') = \sigma_p^2 \exp\left(-\frac{(x-x')^2}{2l^2}\right)$
- Linear Kernel: $k(x, x') = lxx'$
- Cosine Kernel: $k(x, x') = \cos(2\kappa|x - x'|)$
- Exponential Kernel: $k(x, x') = \exp\left(-\frac{|x-x'|}{l}\right)$

Please match each of the plots below to the corresponding kernel name.

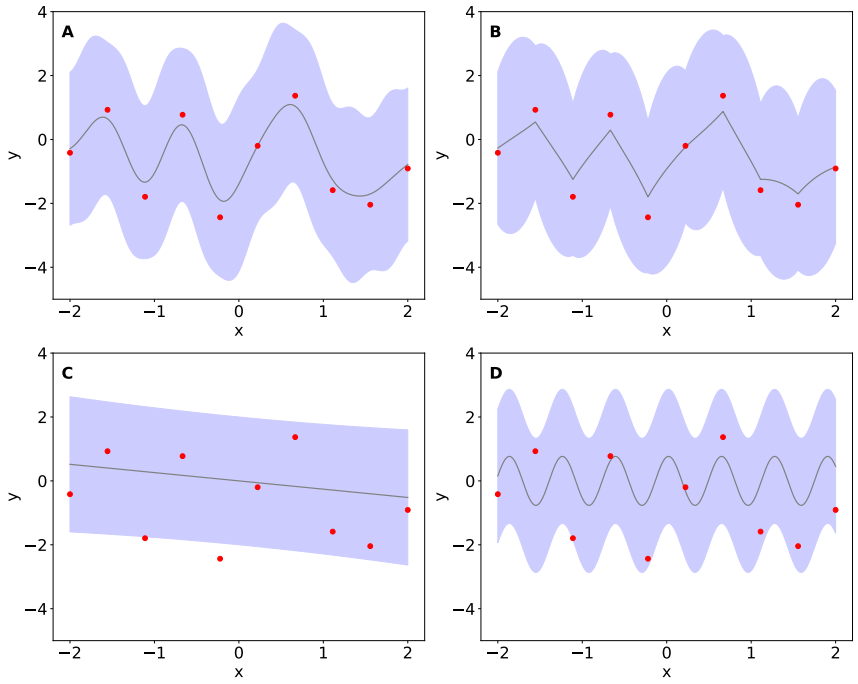


Figure 1: Posterior mean and 95% confidence regions around the mean for different kernels k.

Question 9 *(1 point)*
Which plot corresponds to the linear kernel?

☐ A ☐ B ☐ C ☐ D

Solution:
C. Linear kernel yields a linear posterior mean.

Question 10 *(1 point)*
Which plot corresponds to the cosine kernel?

A A B B C C D D

Solution:
D. The cosine kernel yields a periodic posterior mean.

Question 11 *(1 point)*
Which plot corresponds to the RBF kernel?

A A B B C C D D

Solution:
A. The RBF kernel yields a smoother posterior mean.

Question 12 *(1 point)*
Which plot corresponds to the exponential kernel?

A A B B C C D D

Solution:
B. The exponential kernel yields a rougher posterior mean compared to the setting when the Gaussian kernel is used.

2.2 Kernel Hyperparameters

For the following questions, assume we use a zero-mean Gaussian Process $GP(0, k)$ with the RBF kernel $k(\boldsymbol{x}, \boldsymbol{x}') = \sigma_p^2 \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{x}'\|_2^2}{2l^2}\right)$ with $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^d$ for some $d \in \mathbb{N}_+$, and $\sigma_p^2, l > 0$. Indicate whether the following statements are true or false.

Question 13 *(1 point)*
For any two points $\boldsymbol{x} \neq \boldsymbol{x}'$, the value of $\mathbb{E}_{f\sim GP(0,k)}\left[f(\boldsymbol{x})f(\boldsymbol{x}')\right]$ decreases as l increases.

A True
B False

Solution:
False. $\mathbb{E}_{f\sim GP(0,k)}\left[f(\boldsymbol{x})f(\boldsymbol{x}')\right] = k(\boldsymbol{x}, \boldsymbol{x}')$. For two points $\boldsymbol{x} \neq \boldsymbol{x}'$, as $l \searrow 0$, $k(\boldsymbol{x}, \boldsymbol{x}') \searrow 0$. As $l \nearrow \infty, k(\boldsymbol{x}, \boldsymbol{x}') \nearrow \sigma_p^2$.

Question 14 *(1 point)*
If σ_p^2 increases, the value of $\mathbb{E}_{f\sim GP(0,k)}[f(\boldsymbol{x})^2]$ increases for all $\boldsymbol{x} \in \mathbb{R}^d$.

A True
B False

Solution:
True. Please refer to Lecture 4 slides, page 8.

Question 15 *(1 point)*
For any two points $\boldsymbol{x} \neq \boldsymbol{x}'$ the value of $\mathbb{E}_{f\sim GP(0,k)}\left[(f(\boldsymbol{x}) - f(\boldsymbol{x}'))^2\right]$ decreases as l increases?

A True
B False

Solution:
True. We have $\mathbb{E}_{f\sim GP(0,k)}\left[(f(\boldsymbol{x}) - f(\boldsymbol{x}'))^2\right] = 2\sigma_p^2 - 2k(\boldsymbol{x}, \boldsymbol{x}')$. As l increases, for any $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^d$, $k(\boldsymbol{x}, \boldsymbol{x}')$ increases. Hence, $\mathbb{E}_{f\sim GP(0,k)}\left[(f(\boldsymbol{x}) - f(\boldsymbol{x}'))^2\right]$ decreases as l increases.

2.3 Short Questions

Question 16 *(1 point)*
The posterior distribution $p(f(\boldsymbol{x})|\mathcal{D})$ of a Gaussian process is Gaussian only for specific choices of the kernel k.

A True
B False

Solution:
False. By definition, the posterior distribution is always Gaussian regardless of the kernel function used. More details in Lecture 3 slides, page 13.

Question 17 *(1 point)*
Epistemic uncertainty represents the uncertainty about f due to limited data with respect to a given model.

A True
B False

Solution:
True. This follows from the definition on Lecture 3, Slide 4.

Question 18 *(2 points)*
The function $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by $k(x, x') = (x - x')^2$ is a valid kernel function.

A True

B False

Solution:
False. A valid kernel should be symmetric and positive definite. The above kernel is symmetric but not positive definite. This can be verified by taking any two points $x, x' \in \mathbb{R}$ and computing the determinant of the kernel matrix:

$$\det(\mathbf{K}) = \det\left(\begin{bmatrix} 0 & (x-x')^2 \\ (x'-x)^2 & 0 \end{bmatrix}\right) = -(x-x')^4 \leq 0\,.$$

Simpler observation: for a valid kernel $k(x, x') \leq \sqrt{k(x, x)k(x', x')}$. For the kernel at hand the RHS is 0 but the LHS is not.

Question 19 *(1 point)*
The kernel $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by $k(x, x') = 3e^{(x-x')^2} + 5\cos(x - x')$ is stationary but not isotropic.

A True

B False

Solution:
False. A kernel $k(x, x')$ is stationary if $k(x, x') = k(x - x')$. A kernel $k(x, x')$ is isotropic if $k(x, x') = k(\|x - x'\|)$. We have $(x - x')^2 = |x - x'|^2$ and $\cos(x - x') = \cos(|x - x'|)$), hence k is both stationary and isotropic. Definitions of stationary and isotropic kernel are on Lecture 3 slides, page 23.

Question 20 *(1 point)*
The Random Fourier Features technique constructs a finite-dimensional feature map $\phi(x)$ so that $\hat{k}(x, x') := \phi(x)^\top \phi(x')$ approximates a shift-invariant covariance function $k(x, x')$ of a GP.

A True

B False

Solution:
True. Follows from Lecture 4 slides, Page 23.

2.4 Neural Networks and Gaussian Processes

Consider a fully connected neural network $f : \mathbb{R}^d \to \mathbb{R}$ with a single hidden layer of size n. Here $n, d \in \mathbb{N}$. Specifically, $f(\boldsymbol{x}) = z^{(2)}(\boldsymbol{z}^{(1)}(\boldsymbol{x}))$ where $\boldsymbol{z}^{(1)} : \mathbb{R}^d \to \mathbb{R}^n$ and $z^{(2)} : \mathbb{R}^n \to \mathbb{R}$ are

$$z^{(2)}(\boldsymbol{u}) = b + \boldsymbol{w}^\top \boldsymbol{u}, \qquad \boldsymbol{z}^{(1)}(\boldsymbol{x}) = (\boldsymbol{z}_1^{(1)}(\boldsymbol{x}), \ldots, \boldsymbol{z}_n^{(1)}(\boldsymbol{x}))^\top,$$

where the weights $\boldsymbol{w} = (w_1, \ldots, w_n)^\top \in \mathbb{R}^n$ and the bias $b \in \mathbb{R}$ form the parameters of the neural network. Assume that $\boldsymbol{z}_i^{(1)}(\boldsymbol{x})$ are jointly Gaussian processes and that for all $1 \leq i \leq n$ and $1 \leq j \leq n$ it holds that

$$\mathbb{E}\left[\boldsymbol{z}_i^{(1)}(\boldsymbol{x})\right] = 0 \qquad \mathbb{E}\left[\boldsymbol{z}_i^{(1)}(\boldsymbol{x})\boldsymbol{z}_j^{(1)}(\boldsymbol{x}')\right] = \begin{cases} C(\boldsymbol{x}, \boldsymbol{x}'), & \text{if i = j} \\ 0, & \text{otherwise} \end{cases}. \tag{13}$$

The expectations defined above are taken over the randomness of the Gaussian process. We now wish to analyze the network output $f(\boldsymbol{x})$ as a Gaussian process. For this analysis, we assume that $\boldsymbol{w}$ is deterministic and $b \sim \mathcal{N}(0, \sigma_b^2)$. Moreover, we assume b is independent of $\boldsymbol{z}^{(1)}(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathbb{R}^d$.

Question 21 *(4 points)*
What are the values of the expectation $\mathbb{E}[f(\boldsymbol{x})]$ and covariance $\mathrm{Cov}[f(\boldsymbol{x}), f(\boldsymbol{x}')]$ for two arbitrary points $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^d$?

A $\mathbb{E}[f(\boldsymbol{x})] = 0, \quad \mathrm{Cov}[f(\boldsymbol{x}), f(\boldsymbol{x}')] = \sigma_b^2 + C(\boldsymbol{x}, \boldsymbol{x}')$

B $\mathbb{E}[f(\boldsymbol{x})] = \sum_{i=1}^n w_i, \quad \mathrm{Cov}[f(\boldsymbol{x}), f(\boldsymbol{x}')] = \sigma_b^2 + C(\boldsymbol{x}, \boldsymbol{x}') \sum_{i=1}^n w_i^2$

C $\mathbb{E}[f(\boldsymbol{x})] = 0, \quad \mathrm{Cov}[f(\boldsymbol{x}), f(\boldsymbol{x}')] = \sigma_b^2 + C(\boldsymbol{x}, \boldsymbol{x}') \sum_{i=1}^n w_i^2$

D $\mathbb{E}[f(\boldsymbol{x})] = \sum_{i=1}^n w_i, \quad \mathrm{Cov}[f(\boldsymbol{x}), f(\boldsymbol{x}')] = \sigma_b^2 + C(\boldsymbol{x}, \boldsymbol{x}')$

Solution:
C.

$$\mathbb{E}[f(\boldsymbol{x})] = \mathbb{E}[b] + \sum_{i=1}^n w_i \mathbb{E}\left[\boldsymbol{z}_i^{(1)}(\boldsymbol{x})\right] = 0$$

$$\begin{aligned}\mathbf{Cov}[f(\boldsymbol{x}), f(\boldsymbol{x}')] &= \mathbb{E}\left[\left(b + \boldsymbol{w}^\top \boldsymbol{z}^{(1)}(\boldsymbol{x})\right)\left(b + \boldsymbol{w}^\top \boldsymbol{z}^{(1)}(\boldsymbol{x}')\right)\right] \\ &= \mathbb{E}\left[b^2\right] + \sum_{i=1}^n \sum_{j=1}^n w_i w_j \mathbb{E}\left[\boldsymbol{z}_i^{(1)}(\boldsymbol{x})\boldsymbol{z}_j^{(1)}(\boldsymbol{x}')\right] + \sum_{i=1}^n \mathbb{E}\left[b w_i (\boldsymbol{z}_i^{(1)}(\boldsymbol{x}) + \boldsymbol{z}_i^{(1)}(\boldsymbol{x}'))\right] \\ &= \sigma_b^2 + \sum_{i=1}^n w_i^2 \mathbb{E}\left[\boldsymbol{z}_i^{(1)}(\boldsymbol{x})\boldsymbol{z}_i^{(1)}(\boldsymbol{x}')\right] = \sigma_b^2 + C(\boldsymbol{x}, \boldsymbol{x}') \sum_{i=1}^n w_i^2\end{aligned}$$

Question 22 *(3 points)*
What is the value of the variance $\mathrm{Var}[f(\boldsymbol{x}) - f(\boldsymbol{x}')]$ for two arbitrary points $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^d$?

A $\sum_{i=1}^n w_i^2 \big(C(\boldsymbol{x}, \boldsymbol{x}) + C(\boldsymbol{x}', \boldsymbol{x}') - 2C(\boldsymbol{x}, \boldsymbol{x}')\big)$

B $C(\boldsymbol{x}, \boldsymbol{x}) + C(\boldsymbol{x}', \boldsymbol{x}') - 2C(\boldsymbol{x}, \boldsymbol{x}')$

C $\sum_{i=1}^n w_i^2 \big(C(\boldsymbol{x}, \boldsymbol{x}) + C(\boldsymbol{x}', \boldsymbol{x}')\big)$

D $C(\boldsymbol{x}, \boldsymbol{x}')$

Solution:
Correct solution: **A.**

$$\begin{aligned}
\mathbf{Var}\left[f(\boldsymbol{x}) - f(\boldsymbol{x}')\right] &= \mathbf{Var}\left[f(\boldsymbol{x})\right] + \mathbf{Var}\left[f(\boldsymbol{x}')\right] - 2\mathbf{Cov}\left[f(\boldsymbol{x}), f(\boldsymbol{x}')\right] \\
&= C(\boldsymbol{x}, \boldsymbol{x}) \sum_{i=1}^{n} w_i^2 + C(\boldsymbol{x}', \boldsymbol{x}') \sum_{i=1}^{n} w_i^2 - 2C(\boldsymbol{x}, \boldsymbol{x}') \sum_{i=1}^{n} w_i^2 \\
&= \sum_{i=1}^{n} w_i^2 \big(C(\boldsymbol{x}, \boldsymbol{x}) + C(\boldsymbol{x}', \boldsymbol{x}') - 2C(\boldsymbol{x}, \boldsymbol{x}')\big)
\end{aligned}$$

3 Variational Inference and MCMC

Question 23 *(1 point)* Let $p(w|\mathbf{X}, \boldsymbol{y}) := \frac{1}{Z} p(w) \prod_{i=1}^{n} p(y_i|\boldsymbol{x}_i, w)$ be the posterior distribution after observing the data $(\mathbf{X}, \boldsymbol{y}) := (\boldsymbol{x}_i, y_i)_{i=1:n}$, where $p(w)$ is the prior distribution on the parameter w and Z is a normalization constant. The computation of the mode,

$$w^* := \arg\max_w p(w|\mathbf{X}, \boldsymbol{y}),$$

always requires knowing the value of Z.

A True

B False

Solution:
False. Because Z is just a constant and w^* can be found via, for instance, gradient descent without knowing Z.

Question 24 *(1 point)* Let p be a distribution over $\mathbb{R}$ we wish to approximate, and let $\mathcal{Q}$ be the family of all univariate Gaussian distributions. Consider variational inference with respect to $\mathcal{Q}$:

$$q^* \in \arg\min_{q \in \mathcal{Q}} KL(p||q).$$

Then, q^* is always also the Laplace approximation of p.

A True

B False

Solution:
False. The result of variational inference with respect to $\mathcal{Q}$ is a Gaussian with matching first and second moments of p (see slide 12 of Tutorial 5), whereas the Laplace approximation is a Gaussian with matching **mode** and the second derivative of minus log density of p. In general, the mean is different from the mode, and so is the second moment from the second derivative.

Question 25 *(2 points)* Let p, q, r be arbitrary probability distributions. Which of the following statements about the KL divergence below is true?

A It satisfies the triangle inequality: $KL(q||p) \leq KL(q||r) + KL(r||p)$

B It is symmetric: $KL(q||p) = KL(p||q)$

C $KL(q||p)$ might take a negative value

D Even if $KL(q||r) < \infty$ and $KL(p||r) < \infty$, the value of $KL(q||p)$ might be infinite

Solution:
D. The KL is always non-negative, but does not satisfy the triangle inequality nor is it symmetric. To show **D**, consider r being the standard Gaussian, q uniform on $[0, 1]$ and p uniform on $[3, 4]$. Then, we have $KL(q||r) < \infty$ and $KL(p||r) < \infty$ and $KL(q||p) = \infty$

Question 26 *(2 points)* Recall the activation function $\text{ReLU}(x) := \max(0, x)$. For any $a \in \mathbb{R}$, define

$$\mu_a := \mathbb{E}_{\varepsilon \sim \mathcal{N}(0,1)} \left[\varepsilon \mathbf{1}_{(a,\infty)}(\varepsilon) \right]$$

where $\mathbf{1}_{(a,\infty)}$ is the indicator function of the set (a, ∞), meaning that $\mathbf{1}_{(a,\infty)}(x) = 1$ if $x > a$ and $\mathbf{1}_{(a,\infty)}(x) = 0$ otherwise. Let $w > 0$. Using the reparametrization trick, compute, for any $\sigma > 0$,

$$\nabla_\sigma \mathbb{E}_{x \sim \mathcal{N}(0,\sigma^2)} \left[\text{ReLU}(wx) \right]$$

as a function of μ_a. What is the result?

A $w\sigma\mu_w$
B $w\sigma\mu_\sigma$
C $w\mu_\sigma$
D $w(1 - \mu_\sigma)$
E $w\mu_0$

Solution:
E. For any $x \sim \mathcal{N}(0, \sigma^2)$, we can reparametrize as $x = \sigma\varepsilon$, so that

$$\begin{aligned} \nabla_\sigma \mathbb{E}_{x \sim \mathcal{N}(0,\sigma^2)} \left[\text{ReLU}(wx) \right] &= \mathbb{E}_{\varepsilon \sim \mathcal{N}(0,1)} \left[\nabla_\sigma \text{ReLU}(w\sigma\varepsilon) \right] \\ &= w \cdot \mathbb{E}_{\varepsilon \sim \mathcal{N}(0,1)} \left[\varepsilon \mathbf{1}_{(0,\infty)}(\varepsilon) \right] = w\mu_0. \end{aligned}$$

Question 27 *(4 point)* Let p be the uniform distribution on $[-0.5, 0.5]$. Which one of the following statements about the ELBO is true?

A Let $\mathcal{Q}$ be the family of all Gaussian distributions with mean 0 and variance $\sigma > 0$. Consider the ELBO approximation:

$$\min_{q \in \mathcal{Q}} KL(q||p).$$

Then there exists a $q^* \in \mathcal{Q}$ such that $KL(q^*||p) < \infty$ and q^* solves the above objective

B Let $\mathcal{Q}$ be the family of all Gaussian distributions with mean 0 and variance $\sigma > 0$. Consider an ELBO approximation where the order of p and q in the KL exchanged, i.e.:

$$\min_{q \in \mathcal{Q}} KL(p||q).$$

Then there exists a $q \in \mathcal{Q}$ such that $KL(p||q) = \infty$

C Let $\mathcal{Q}$ be the family of all Gaussian distributions with variance 1. Consider the ELBO approximation:

$$\min_{q \in \mathcal{Q}} KL(q||p).$$

Then there exists a $q^* \in \mathcal{Q}$ such that $KL(q^*||p) < \infty$ and q^* solves the above objective

D Let $\mathcal{Q}$ be the family of all Gaussian distributions with variance 1. Consider the ELBO approximation where the order of p and q in the KL exchanged, i.e.:

$$\min_{q \in \mathcal{Q}} KL(p||q).$$

Then there exists a $q^* \in \mathcal{Q}$ such that $KL(p||q^*) < \infty$ and q^* solves the above objective

Solution:
D. For A and C, the objective is always ∞ for any $q \in \mathcal{Q}$. For B, the KL divergence can be computed as:

$$KL(p||q) = \int_{-0.5}^{0.5} -\log q(x)dx = c + \frac{1}{2}\log 2\pi\sigma^2 + \frac{1}{\sigma^2}\int_{-0.5}^{0.5} \frac{x^2}{2}dx$$

where c is a universal constant and σ^2 is the variance of q. This objective is clearly finite for all $\sigma > 0$. For D, we can compute the KL divergence as

$$KL(p||q) = \int_{-0.5}^{0.5} -\log q(x)dx = c + \int_{-0.5}^{0.5} \frac{(x-\mu)^2}{2}dx = c + \frac{1}{6}\Big((0.5-\mu)^3 - (-0.5-\mu)^3\Big)$$

where c is a universal constant. This can be further simplified to:

$$KL(p||q) = c + \frac{1}{6}\left(\frac{1}{4} + 3\mu^2\right)$$

which attains the minimum at $\mu = 0$.

Question 28 *(3 point)* Consider an ergodic Markov chain $\{X_N\}_{N\geq 1}$ with a finite number of states. Let Λ be the probability transition matrix of the Markov chain. Which one of the following statements is **false**?

A For every state x, $\lim_{N\to\infty} P(X_N = x) > 0$

B No row of Λ can contain the number 1

C The stationary distribution of $\{X_N\}_{N\geq 1}$ does not depend on X_1

D A matrix is called *doubly-stochastic* if the sum of every row and column is 1. If the transition matrix Λ is doubly-stochastic, then the stationary distribution is the uniform distribution over the states

Solution:
B. Item **A** is true because of Slide 25, lecture 6. Item **B** is false: see the example on Slide 24 of lecture 6. Item **C** is true; see Slide 25 of lecture 6. Item **D** is true; this can be seen by simply noticing that the uniform distribution $\pi_i = \frac{1}{n}$, where n is the number of states, is a solution to the equation of the stationary distribution $\pi^\top \Lambda = \pi^\top$.

4 Calibration of uncertainty estimates

Suppose we have trained a neural network model $f_{\boldsymbol{\theta}}(\cdot)$ on a binary classification dataset. In particular, given a four-dimensional feature vector $\boldsymbol{x} \in \mathbb{R}^4$, our model predicts the probability of the target $y \in \{0, 1\}$ by feeding the scalar output of the neural network through a sigmoid function $\sigma(u) = 1/(1 + e^{-u})$, i.e.,

$$\hat{p}(y = 1|\boldsymbol{x}) := \sigma\left(f_{\boldsymbol{\theta}}(\boldsymbol{x})\right) \ .$$

Now, we perform an evaluation experiment for our model on 6 new validation samples. For each sample $n = 1, ..., 6$, we note down the input vector $\boldsymbol{x}_n$, the output of the last neural network layer $z_n := f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)$, the predicted output $\hat{p}(y_n = 1|\boldsymbol{x}_n)$, and the ground-truth label y_n. Table 1 summarizes the results of our experiment.

n	$\boldsymbol{x}_n$	$z_n = f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)$	$\hat{p}(y_n = 1\|\boldsymbol{x}_n)$	y_n
1	[-5.027, 0.259, -2.194, 3.952]	-2.197	0.1	0
2	[-1.376, -0.355, 0.698, 1.648]	-0.847	0.3	1
3	[0.909, -2.306, 1.348, 1.673]	0.405	0.6	1
4	[0.849, 1.575, 2.739, 0.837]	0.405	0.6	1
5	[0.374, 0.530, 0.358, 4.141]	1.386	0.8	1
6	[-2.381, -0.427, 1.633, -3.093]	1.386	0.8	1

Table 1: Summary of evaluation experiment.

Question 29 *(3 points)* What is the expected calibration error (ECE) of the model? Use three linearly spaced interval bins (each of size $\frac{1}{3}$) to group the validation samples according to their respective $\hat{p}(y_n = 1|\boldsymbol{x}_n)$ for the computation of the ECE!

A 0.2
B 0.3
C 0
D 0.75
E 0.15

Solution:
B. We group the samples in three bins with separation thresholds at $\frac{1}{3}$ and $\frac{2}{3}$ according to their predicted output. The first bin contains samples 1 and 2, the second contains 3 and 4, and the third contains 5 and 6. We calculate the relative frequency of positive samples for each bin. We obtain 0.5, 1 and 1 respectively as well as the mean predictive confidence of the positive class for each bin $(0.2, 0.6, 0.8)$. We then compute the absolute difference between the two for each bin $(0.3, 0.4, 0.2)$ and compute an average weighted by the relative number of validation samples within a respective bin. The ECE is 0.3.

Question 30 *(2 point)* Which of the following statements agrees with the results of the evaluation experiment?

A Depending on the predicted confidence (i.e., the bin), the model *varies* between being *overconfident* and *underconfident* that $y = 1$.
B The model tends to be *underconfident* that $y = 1$ within all considered bins.
C The model tends to be *overconfident* that $y = 1$ within all considered bins.



D The model is *perfectly calibrated*.

Solution:
B. The computed mean predictive confidence of the positive class is smaller than the relative occurrence of positive labels within all three bins. Thus, the model is underconfident with respect to the occurrence of an event for all considered bins.

Question 31 *(2 point)* We can improve the calibration of our network by applying a linear scaling on the logits of our network,

$$\hat{p}(y = 1|\boldsymbol{x}) = \sigma\left(b f_{\boldsymbol{\theta}}(\boldsymbol{x}_n) + a\right),$$

where $a \in \mathbb{R}$ and $b \in \mathbb{R}$ are parameters chosen in order to minimize the ECE on the validation dataset. How would the confidence of the model that an event takes place (i.e., $\hat{p}(y = 1|\boldsymbol{x})$) change with the linear scaling using $a > 0$ and $b = 1$?

A The *confidence* of the model that $y = 1$ would *increase*.

B The *confidence* of the model that $y = 1$ would *decrease*.

C The *confidence* of the model that $y = 1$ would remain *the same*.

D The *confidence* of the model that $y = 1$ would only *decrease* for samples $\boldsymbol{x}_n$ for which z_n *is negative*.

Solution:
A. The correct choice is A. Since the sigmoid function is monotonically increasing, by adding a positive constant offset to each z_n the predicted output $\hat{p}(y_n = 1|\boldsymbol{x}_n)$ (i.e., the confidence that an event takes place) increases for each input vector $\boldsymbol{x}_n$.

5 Bayesian optimization

In the following Bayesian optimization problem, we are interested in finding the maximum of a function f drawn from a Gaussian process (GP) with zero mean and known kernel k. We observe t noisy observations $S_t = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^t$ that satisfy $y_i = f(\boldsymbol{x}_i) + \varepsilon_i$ where $\varepsilon_1, \ldots, \varepsilon_t \sim \mathcal{N}(0, \nu^2)$ is i.i.d. Gaussian noise with variance ν^2. By $\mu_t(\boldsymbol{x})$ and $\sigma_t(\boldsymbol{x})$, we denote the posterior mean and standard deviation of f at $(\boldsymbol{x})$, after observing t points $(\boldsymbol{x}_i, y_i)$. To select the next query point $\boldsymbol{x}_{t+1}$, we use the *acquisition function*

$$a_t(\boldsymbol{x}) = (\mu_t(\boldsymbol{x}) - f_t^* - \zeta)\Phi\left(\frac{\mu_t(\boldsymbol{x}) - f_t^* - \zeta}{\sigma_t(\boldsymbol{x})}\right) + \sigma_t(\boldsymbol{x})\phi\left(\frac{\mu_t(\boldsymbol{x}) - f_t^* - \zeta}{\sigma_t(\boldsymbol{x})}\right),$$

where $\zeta \in \mathbb{R}$ is a chosen hyperparameter and f_t^* designates the maximum of the observed function values across the previously queried points, i.e., $f_t^* := \max_{i=1,\ldots,t} y_i$. Here, ϕ is the probability density function ($\phi(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}}$) and Φ is the cumulative distribution function ($\Phi(u) = \int_{-\infty}^{u} \phi(z)dz$) of a standard normal variable. Figure 2 shows the contour plot for the acquisition function $a_t(\boldsymbol{x})$.

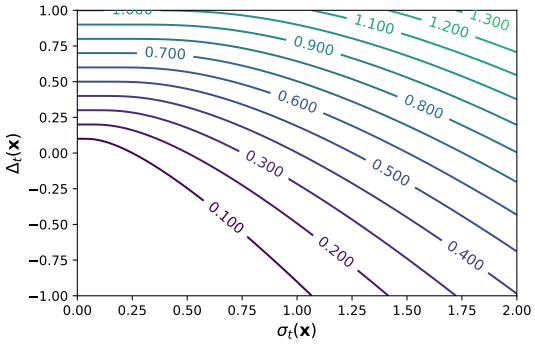


Figure 2: Contour plot for acquisition function $a_t(\boldsymbol{x})$ depending on $\sigma_t(\boldsymbol{x})$ and $\Delta_t(\boldsymbol{x}) = \mu_t(\boldsymbol{x}) - f_t^* - \zeta$.

Question 32 *(4 points)* How does the search behavior of the optimization algorithm change for larger values of ζ?

- A ζ does *not* have *any effect* on the search behavior of the algorithm.
- B The algorithm will become *less explorative* and *more exploitative* for larger values of ζ.
- C The algorithm will become *more explorative* and tend to focus on points in regions with larger uncertainty for larger values of ζ.
- D The algorithm will become *less explorative* and *less exploitative* for larger values of ζ.

Solution:
C. Consider two points lying on the same contour line (i.e., with the same acquisition function value). As we increase ζ, $\Delta_t(\boldsymbol{x})$ becomes smaller and thus we shift both points downwards on the $\Delta_t(\boldsymbol{x})$ axis. As a result, the value of the acquisition function at the point with the initially higher variance $\sigma_t(\boldsymbol{x})$ is now greater than at the point with the lower variance. The acquisition function thus more strongly favors points with larger variance (uncertainty) and thus, following the corresponding definition the algorithm becomes more explorative.

Question 33 *(1 point)* An alternative approach to Bayesian Optimization with an acquisition function $a_t(\boldsymbol{x})$ considered above would be Thompson sampling. Indicate whether it is true that for Thompson

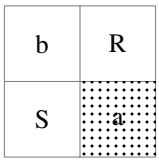
sampling, the next query point $\boldsymbol{x}_{t+1}$ is selected to maximize the function $\tilde{f}$, which is randomly drawn from $GP(\mu_t, k_t)$.

A True B False

Solution:
True. The statement is true (see Lecture 8, Slide 44).

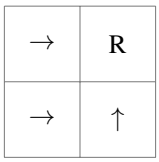
6 Reinforcement Learning

Our goal is to train a rescue robot to navigate from its starting position (S) to the rescue location (R). We model its environment as a 2x2 grid, i.e., 4 states, with S and R as shown below.



In every state, the robot can choose to move up ($\uparrow$), right ($\rightarrow$), left ($\leftarrow$) or down ($\downarrow$). The transitions are deterministic, i.e., the robots will follow the action with probability one. If the robot is at the boundary of the grid, and takes an action that would move it outside the grid, it will stay in the same location. For example, if it takes action $\downarrow$ in state S, it will remain in S. When the robot reaches state R, it receives a terminal reward of 1, i.e., $V^{\pi}(R) = 1, \forall \pi$, and its task is completed. In all other states, the robot does not receive any reward. We use a discount factor of $\gamma = 0.9$.

Question 34 *(4 points)* The model described above is not a perfect description of the system in winter when the streets are icy and slippery. From last year's data, we know that state a (dotted area) was influenced by this. In that state, if the robot picks action $\uparrow$, it moves left with probability 0.4 and up with probability 0.6, all the other transitions remain unchanged (deterministic). Given the policy π_0 shown below, what is its expected discounted return $V^{\pi_0}(S)$?



Note: The solutions are rounded to two digits.

A $V^{\pi_0}(S) \approx 0.72$

B $V^{\pi_0}(S) \approx 0.8$

C $V^{\pi_0}(S) \approx 0.81$

D $V^{\pi_0}(S) \approx 0.9$

E $V^{\pi_0}(S) \approx 0.92$

Solution:
A. We evaluate policy π_0 using the following system of linear equations (see lecture 9, slide 15):

1. $V^{\pi_0}(R) = 1$
2. $V^{\pi_0}(b) = r(b, \rightarrow) + \gamma P(R|b, \rightarrow) V^{\pi_0}(R) = 0.9$
3. $V^{\pi_0}(a) = r(a, \uparrow) + \gamma \left(P(R|a, \uparrow) V^{\pi_0}(R) + P(S|a, \uparrow) V^{\pi_0}(S) \right) = 0.54 + 0.36 V^{\pi_0}(S)$
4. $V^{\pi_0}(S) = r(S, \rightarrow) + \gamma P(a|S, \rightarrow) V^{\pi_0}(a) = 0.9 V^{\pi_0}(a)$.

Solving equations 3 and 4, we obtain $V^{\pi_0}(a) \approx 0.799$ and $V^{\pi_0}(S) \approx 0.719$.
Note: The solutions are rounded to three digits here.

Question 35 *(2 points)*
We want to use policy iteration to compute an optimal policy. If we initialize with π_0, how many iterations do we need to converge to the optimal policy?

A 0, π_0 is already optimal.

B 1

C 2

D 3

Solution:
B. After a single iteration we get π_1, which is the greedy policy with respect to V^{π_0}:

$\rightarrow$	R
$\uparrow$	$\uparrow$

As in the previous question, we evaluate policy π_1 using the following system of linear equations (see lecture 9, slide 15):

1. $V^{\pi_1}(R) = 1$
2. $V^{\pi_1}(b) = r(b, \rightarrow) + \gamma P(R|b, \rightarrow) V^{\pi_1}(R) = 0.9$
3. $V^{\pi_1}(S) = r(S, \uparrow) + \gamma P(b|S, \uparrow) V^{\pi_1}(b) = 0.81$
4. $V^{\pi_1}(a) = r(a, \uparrow) + \gamma \left(P(R|a, \uparrow) V^{\pi_1}(R) + P(S|a, \uparrow) V^{\pi_1}(S) \right) = 0.54 + 0.2916 = 0.8316.$

Using Bellman's theorem (see lecture 10, slide no. 15), one can verify that this is indeed the optimal value function.

Question 36 *(2 points)*
As mentioned earlier, the icy area was inferred from *last* year's data. For this year, we do not know yet where the icy area is going to be. Suppose that this is our environment:

b	R
S	a

We have sampled the following trajectory this year: $(S, \uparrow, b, \rightarrow, S, \uparrow, b, \rightarrow, R)$.
We will now use Q-learning with constant $\alpha = 0.1$ and discount factor $\gamma = 0.9$ to learn the Q-values. Suppose that we initialize $Q(R, a) = 1$, for all $a \in \{\rightarrow, \leftarrow, \uparrow, \downarrow\}$. All other Q-values are initialized to 0. What is the value $Q(S, \rightarrow)$ after we process the trajectory?
Note: The solutions are rounded to two digits.

A 0
B 0.53
C 0.59
D 0.66
E 0.72

Solution:
A. For the Q-learning update, see lecture 10, slide number 30. After processing the trajectory, the only Q-value that gets updated to a non-zero value is:

$$Q(b, \rightarrow) = (1 - \alpha)Q(b, \rightarrow) + \alpha \left(r(b, \rightarrow) + \gamma \max_{a'} Q(R, a') \right) = 0.09. \tag{14}$$

Note that there are two transitions that contain b, the non-zero update happens only in the second transition.

6.1 True/False questions

Question 37 *(1 point)*
For an MDP with $n < \infty$ states, $m < \infty$ actions, and discount factor $\gamma \in [0, 1)$, policy iteration converges to an optimal policy in a number of iterations that is polynomial in n and m.

A True
B False

Solution:
True. See lecture 9, p. 16: "Guaranteed to converge to an optimal policy p in $O(n^2 m/(1 - \gamma))$ iterations", where n is the number of states, m is the number of actions, and γ is the discount factor.

Question 38 *(1 point)*
Q-learning is an off-policy method.

A True

B False

Solution:
True. Lecture slides 11, p. 8: We only need that $(\alpha_t)_{t\in\mathbb{N}}$ satisfies $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$, and all state-action pairs are visited infinitely often, but there's no assumption that the sampling and the updated policy need to be the same. Thus, it is an off-policy method.

Question 39 *(1 point)*
There exists an MDP with a finite number of states and actions, and a discount factor $\gamma \in [0, 1)$, for which the value iteration does not converge.

A True

B False

Solution:
False. The Bellman operator B is a contraction, so we have the following property: $\|BV_{t+1} - BV_t\| \leq \gamma\|V_{t+1} - V_t\|$ (see lecture slides 9, p.21). Since $\gamma < 1$, value iteration does converge.

Question 40 *(1 point)*
Every MDP with finitely many states and actions possesses a deterministic optimal policy.

A True
B False

Solution:
True. This is a direct implication from Bellman's optimality theorem (see slides 11, p.2): For an optimal policy π^*, the following condition holds:

$$V^{\pi^*}(x) = \max_a \left[r(x,a) + \gamma \sum_{x'} P(x'|x,a) \cdot V^{\pi^*}(x') \right], \quad \forall x \tag{15}$$

In fact, the Bellman optimality theorem can also be used to construct an optimal, deterministic policy from any optimal, stochastic policy.

Question 41 *(1 point)*
The Q-learning algorithm estimates a transition model with the previously collected transitions.

A True
B False

Solution:
False. Q-learning is a model-free algorithm. It learns the Q-function directly and does not try to learn the transition model. See lecture 10, slides 28 and 29.

Question 42 *(1 point)*
For large or continuous state spaces, computing the Q-values for every (x, a) is infeasible. A common approach is to parametrize the Q-function as $Q(x, a; \theta)$ with parameters θ. For every transition (x, a, r, x'), the basic Q-learning updates θ with respect to a loss function $l(\theta; x, a, r, x')$. This basic algorithm is typically rather slow. DQN (Deep Q-Network) aims to accelerate the learning process by using a loss function $L(\theta; \mathcal{D}) = \sum_{(x,a,r,x') \in \mathcal{D}} l(\theta; x, a, r, x')$ that depends on the dataset $\mathcal{D}$ rather than on a single transition.

A True
B False

Solution:
True. See lecture 11, slides no. 17 and 18: to update the parameters θ, the basic Q-learning algorithm (p. 17) aims to minimize the loss function only for the latest transition. Instead, DQN (Deep Q-Network) aims to minimize the cumulative loss for all transitions observed so far (p.18).

Question 43 *(1 points)*
Recall that policy gradient methods rely on baselines to reduce the variance of the gradient. As discussed in the lecture, the gradient remains unbiased when adding a baseline, i.e. for every b: $\nabla J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[r(\tau) \nabla \log \pi_\theta(\tau)] = \mathbb{E}_{\tau \sim \pi_\theta}[(r(\tau) - b) \nabla \log \pi_\theta(\tau)]$. Any constant baseline b will always reduce the variance.

A True

B False

Solution:
False. In homework 6, question 4, we showed that if $b^2 \leq 2b \cdot r(x, a)$, for all (x, a), then b does reduce the variance (or keep it the same). If b doesn't satisfy this condition, the variance can potentially increase.



+1/25/35+

Answer Sheet of the Probabilistic Artificial Intelligence 2022/23 Exam

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

⟵ Please encode your student number on the left, and write your first and last names below.

Firstname and Lastname:

. . . . . .

Question 1: A B C D
Question 2: A B C D
Question 3: A B C D E
Question 4: A B C D
Question 5: A B
Question 6: A B
Question 7: A B
Question 8: A B
Question 9: A B C D
Question 10: A B C D
Question 11: A B C D
Question 12: A B C D
Question 13: A B
Question 14: A B
Question 15: A B
Question 16: A B
Question 17: A B
Question 18: A B
Question 19: A B
Question 20: A B
Question 21: A B C D
Question 22: A B C D
Question 23: A B
Question 24: A B
Question 25: A B C D
Question 26: A B C D E
Question 27: A B C D
Question 28: A B C D
Question 29: A B C D E
Question 30: A B C D
Question 31: A B C D
Question 32: A B C D
Question 33: A B
Question 34: A B C D E
Question 35: A B C D
Question 36: A B C D E
Question 37: A B
Question 38: A B
Question 39: A B
Question 40: A B
Question 41: A B
Question 42: A B
Question 43: A B